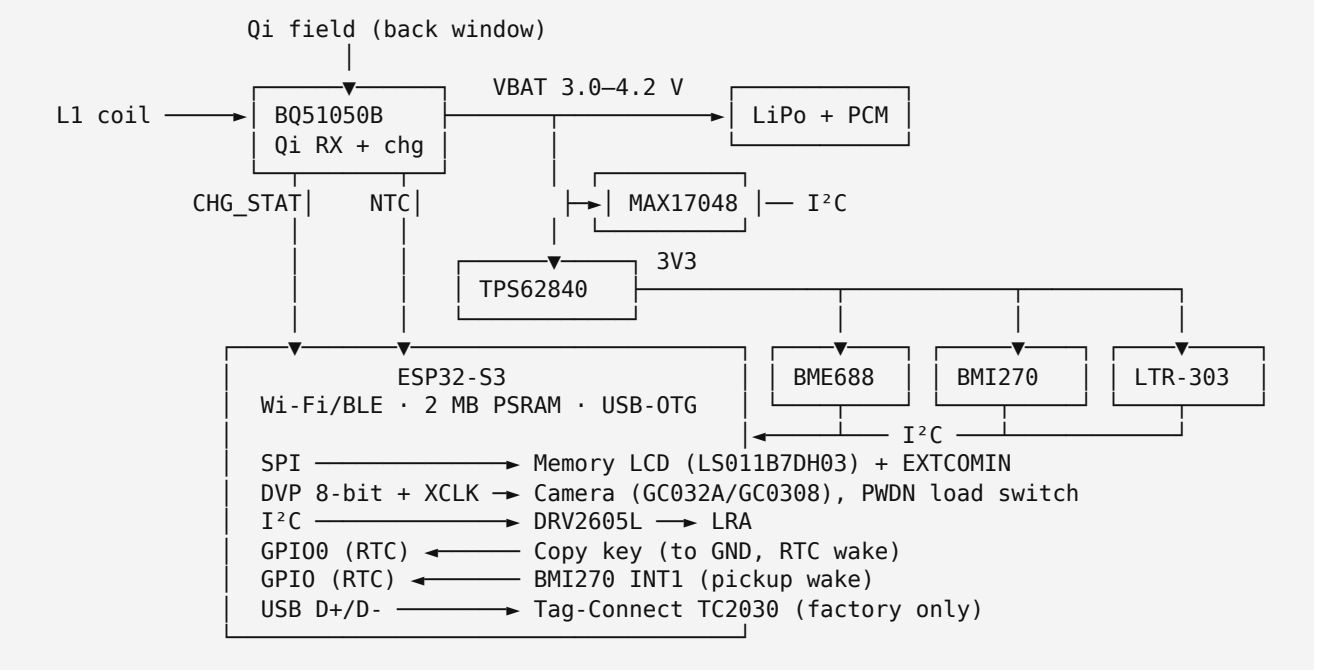
04 · Electrical Design

Doc: LOT-HW-01-ELEC · **Rev:** A (2026-11-14)

1. BLOCK DIAGRAM



2. PIN MAP (ESP32-S3, EVT)

Function	GPIO	Notes
I²C SDA / SCL (sensors, haptic, gauge)	8 / 9	400 kHz, 4.7 kΩ pull-ups on the 3V3 sensor rail
LCD SCLK / MOSI / CS	12 / 11 / 10	Memory LCD: CS is active-high, LSB-first
LCD EXTCOMIN	13	1 Hz toggle from the ULP/LEDC during sleep (VCOM inversion)
LCD DISP	14	Display on/off
CAM D0–D7	15,16,17,18,39,40,41,42	DVP
CAM PCLK / VSYNC / HREF / XCLK	4 / 5 / 6 / 7	XCLK 20 MHz from LEDC
CAM SIOD / SIOC	shares I²C 8 / 9	SCCB-compatible
CAM PWDN (load switch EN)	21	Camera fully unpowered when idle
Copy key	0	RTC GPIO wake, also BOOT strap. Hold at reset = download mode
IMU INT1	1	RTC wake on pickup
Qi CHG_STAT / PGOOD	2 / 3	Charging UI, and ext0 wake on dock
Fuel gauge ALRT	47	Low-battery interrupt
USB D- / D+	19 / 20	Native USB-Serial/JTAG for provisioning (`lotctl`)

3. POWER BUDGET AT THE DEFAULT 5-MINUTE CADENCE

State	Current	Time per hour	mAh per hour
Deep sleep (RTC + ULP EXTCOMIN + sensors idle + LCD static)	18 µA	3 590 s	0.018
Wi-Fi wake, fetch (fast reconnect with a cached BSSID/channel, TLS session resumption)	95 mA avg	12 × 0.9 s	0.285
BME688 forced-mode read, every 15 min	1 mA	4 × 1 s	0.001
Notification: haptic 300 ms + LCD write	80 mA	~3 × 0.4 s	0.027
Copy press: wake + POST	95 mA	~2 × 1.2 s	0.063
Total			**≈ 0.39 mAh per hour ≈ 9.5 mAh per day**

- EVT cell, 150 mAh × 0.85 usable ≈ **13 days** in the ideal case. Plan on **≥ 5 days** with real Wi-Fi retries.
- DVT cell, 70 mAh ≈ **6 days** ideal, ≥ 2–3 days real.
- A persistent socket for true push (v2) costs about 2–3 mA average with DTIM-3 modem sleep, which is about 1 day on the DVT cell. That is why v1 polls.
- Adaptive cadence: 2 minutes while the user is active on lot-systems.com (`next_poll_s` comes from the server), 30 minutes overnight per the user's circadian phase.

4. CHARGING

- **BQ51050B:** WPC 1.1 Qi receiver with an integrated Li-ion charger (4.20 V, set the ILIM resistor to about 150 mA = 1 C for the EVT cell and 70 mA for DVT). The TS/CTRL pin is wired to a 10 kΩ NTC on the cell.
- **Charger accessory:** EVT ships with an off-the-shelf WPC-certified 5 W pad. PVT ships with a LOT-branded stainless puck that has a zirconia top, a magnet ring for alignment (the steel walls of the device are only weakly attracted, so use a thin steel shim plate behind GL2) and USB-C input.
- The screen shows `Charging. 64%` and then `Charged.` (LOT style, with a period).

5. SENSORS: "AI-GRADE, OFF-THE-SHELF"

Sensor	AI feature used	Output to LOT
Bosch BME688	BSEC2 library (IAQ index), plus a custom gas-class model trained in BME AI-Studio (e.g. "coffee brewing", "fresh air", "stale room")	`temp_c, rh, hpa, iaq, gas_class` every 15 min
Bosch BMI270	On-chip wrist-wear and any-motion features, used for pickup-to-wake and double-tap = dismiss	Wake events only. No raw motion leaves the device
Lite-On LTR-303	Lux used for a bedtime "quiet" window	`lux` (coarse) with each telemetry packet

The BME688 gas class gives LOT a context signal: "coffee brewing" detected leads the site to log a coffee moment and suppress the "Coffee time!" nudge.

6. SCHEMATIC AND LAYOUT TASKS (NEXT SESSION)

1. KiCad 8 project `hardware/lot-tile/pcb/`: schematic for all of the above, with ERC clean.
2. Layout: 36 × 36 mm, 4 layers (signal, GND, power, signal), 0.6 mm, ENIG, with cut-outs for the LRA and camera.

3. Antenna keep-out per Espressif's ESP32-S3-MINI-1 hardware design guide, placed on the front-glass edge.
4. Test points for bed-of-nails: VBAT, 3V3, USB, EN, GPIO0, I²C, CHG_STAT.